Power switch board: bring-up

Nine steps, each fitted and tested before the next · 16 October 2026

Each step adds one working part of the circuit and is tested before the next goes in, so a fault shows up in the step that caused it. **Where the board is now:** D1–D8 and U1 are fitted, and nothing else.

Setting up

Feed the board through a resistor. Pack + → the 22 Ω 50 W resistor → P1. Pack − → P3. Use temporary leads; the permanent ones go in at step 9. A dead short then draws about an amp instead of whatever the pack can deliver, so keep the resistor in the pack lead until step 9.

Give the output a load. Any resistor of a few kilohms (4.7 kΩ is fine) soldered into J10's empty pins 1 and 3. Without it the output reading floats when the switch is off.

3S and 6S. 3S is one pack on its own, 11–12.6 V. 6S is both packs through the series harness, 22–25 V, which is what the board runs on. Do every check on 3S first, then on 6S, and write the readings in the yellow columns. A grey dash means that check isn't made at that voltage.

J3 is the button. Shorting its two pins with a wire is the same as pressing the robot's on/off button.

Where to measure

Black meter lead on ground (J10 pin 3, or P3) unless the row says otherwise. U1's pin 1 is the corner next to its notch or dot; from above, the pins count down that side 1–4 and back up the other 5–8.

What	Where
Output	J10 pin 1 (the square pad) to pin 3. Until C5 is fitted, its two empty holes are the same pair, the + hole being the output.
Gate to source	Across R2. The meter may show it negative; read the size.
Current	The voltage across the 22 Ω, divided by 22, in amps: 0.22 V is 10 mA.
Reference	U1 pin 2.
Hold threshold	U1 pin 6.
Enable	U1 pin 1.
Cutoff divider	C3's + lead.

Before you start

Polarised parts. C3, C4 and C7 are tantalum and C5 and C6 are electrolytic. A tantalum fitted backwards burns. C7 was bought as a tantalum where the board expects a film, so its pads carry no + mark: its + goes on the VSRC side, towards J2.

The four TO-92s have two different pinouts. Q3, Q4 and Q5 are BC546B transistors, pins 1 C, 2 B, 3 E. U2 is a TL431 reference, pins 1 REF, 2 A, 3 K. Don't mix them up. The flat face goes to the flat on the silkscreen, and the middle lead bends back to fit the staggered holes.

The capacitor bank stays charged after the switch turns off, from step 4 on. Wait for the output to read under 1 V before handling the board.

Step 1 · Check what's already in

Nothing to fit. Every diode is on a net with nothing else on it yet, so the meter's diode test reads each one cleanly, and this is the last time a reversed diode will be this easy to find. The triangle in the board drawing (last page) is the anode and the bar is the cathode.

Part	Type	Job	Red on anode, black on cathode: 0.5–0.7 V	Probes reversed: OL
D1	1N5245B, 15 V zener	Clamps the gates at 15 V		
D2	1N4148	Lets Q4 pull the gates off		
D3	1N4148	Holds Q4 off while the gates are on		
D4	1N4148	Controller's hold input		
D5	1N4148	SBC's hold input		
D6	1N4148	The button pulls the gates on through it		
D7	1N4148	Lets the controller read the button		
D8	1N4148	Soft start: C1 into the gates		

The zener reads like an ordinary diode in this direction; its 15 V is checked in step 2.

Check	Expect	Result
U1's notch or dot	At the square pad, which is pin 1	
P1 to P3, meter on ohms	Open (OL): F1 isn't fitted yet	
J10 pin 1 to pin 3, meter on ohms	Not a short: anything above a few hundred ohms (U1 is across it)	

If not: a diode that reads the wrong way round is in backwards; turn it round. Each one stands on its cathode pad, band down.

Step 2 · The switch, worked by hand

Part	Value	Note
F1	Keystone 3568 holder, 20 A mini blade fuse	The rating doesn't matter yet: the 22 Ω limits the current
Q1, Q2, Q6, Q7	IRF4905 P-channel MOSFET, TO-220	Tab as the silkscreen shows
R2	1 MΩ	Holds the switch off: gate to source
R3, R4	10 Ω	Gate resistors. R3's body stands over pad 2, the hole without the printed ring, to clear its neighbours
R5	470 kΩ	How hard the button pulls the gates. Don't swap it with R2
J3	JST-XH 2-pin header	The button
C2	100 nF 50 V ceramic	U1's supply capacitor; U1 is powered from this step on

Check

Condition	Measure at	Expect	3S	6S
J3 open	Output	About 0 V		
J3 open	Across the 22 Ω	About 0 V		
J3 open	Gate to source	About 0 V		
J3 shorted	Output	Pack voltage, within a fraction of a volt		
J3 shorted	Across the 22 Ω	Well under 1 V: the current your load draws		
J3 shorted	Gate to source	7–8 V on 3S; 15 V or less on 6S		
J3 opened again	Output	Back to 0 V within a fraction of a second		

If not: gate to source near zero with J3 shorted points at R5, D6 or J3. A healthy gate reading with a dead output points at the MOSFETs or the fuse. More than about 16 V on a full 6S pair means D1 isn't clamping.

Step 3 · Turn-off

Fit C7 on its own first, to see the problem the next three parts solve.

Part	Value	Note
C7	2.2 µF 35 V tantalum, as bought (the board is drawn for a film)	**Polarised, and the board has no + mark for it.** + goes in the hole marked VSRC in the drawing, the one towards J2

Check

Condition	Measure at	Expect	3S	6S
J3 shorted, then opened	Gate to source	Falls away over a few seconds. Expected: nothing pulls the gates off quickly yet		

Then fit

Part	Value	Note
Q4	BC546B NPN, TO-92	Pulls the gates off when the button lets go
R8	47 kΩ	Q4's base
R9	1 MΩ	Q4's base to emitter

Check

Condition	Measure at	Expect	3S	6S
J3 shorted, then opened	Gate to source	To within about 1.5 V of zero at once		
J3 opened	Output	To 0 V at once		

If not: still slow points at Q4 turned round, R8 or D2.

Step 4 · Soft start and the capacitor bank

Part	Value	Note
C1	220 nF 63 V film, 5 mm pitch	The soft start
Q5	BC546B NPN, TO-92	Resets C1
R25	10 kΩ	Q5's base
R26	1 kΩ	Q5's collector. The same part as R18

Check: should behave exactly as at the end of step 3

Condition	Measure at	Expect	3S	6S
J3 shorted	Output	Pack voltage		
J3 opened	Gate to source	To within about 1.5 V of zero at once		

If not: if the switch no longer comes on, look at Q5 (turned round or shorted) and C1.

Then fit

Part	Value	Note
C5, C6	470 µF 35 V low-ESR electrolytic, 10 × 16 mm	The bank. + to the + mark

Check

Condition	Measure at	Expect	3S	6S
J3 shorted	Output	Pack voltage		
J3 opened	Output	Falls over several seconds through the load: the bank emptying, which is expected		
J3 opened	Across the 22 Ω	About 0 V		

Nothing but your load drains the bank. Wait for the output to read under 1 V before handling the board.

Step 5 · The reference

Part	Value	Note
U2	TL431ACLP reference, TO-92	Pins 1 REF, 2 A, 3 K: not the transistors' pinout
R10	15 kΩ	Feeds the reference

Check

Condition	Measure at	Expect	3S	6S
J3 shorted	U1 pin 2	2.47–2.52 V	—	

If not: U2 turned round, or R10.

Step 6 · The cutoff divider

Part	Value	Note
R11	100 kΩ 1 %	Divider top: sets the cutoff, so use a 1 % part
R12	15 kΩ 1 %	Divider bottom: likewise 1 %
C3	22 µF 16 V tantalum	The cutoff's filter. Polarised: + to the + mark
R13	22 kΩ	Comparator input
R14	10 MΩ	Hysteresis
R15	47 kΩ	Pull-up on the enable line

Check

Condition	Measure at	Expect	3S	6S
J3 shorted	C3's + lead	0.130 × the output voltage: 1.56 V at 12.0 V, 3.26 V at 25.0 V		

If not: R11 and R12 swapped, or the wrong values.

Step 7 · The hold

Put a wire link in J3 to hold the board on for this step. For the 3 V, use two AA cells in series or the Banana Pi's 3.3 V pin: + on J2 pin 1, − on J2 pin 3.

Part	Value	Note
R16, R17	100 kΩ	Hold pull-downs: a loose wire means off
R18	1 kΩ	Charges the hold capacitor
C4	10 µF 16 V tantalum	Rides through a controller reset. Polarised
R19	100 kΩ	Hold bleed
R20	22 kΩ	Hold comparator input
R21	10 MΩ	Hold hysteresis
R22, R23	100 kΩ 1 %	Hold threshold: half the reference
J2	JST-XH 4-pin header	The hold lead to the carrier

Check, with the J3 link in

Condition	Measure at	Expect	3S	6S
J3 linked	U1 pin 6	Half of U1 pin 2's reading (the reference from step 5): about 1.25 V	—	
Nothing on J2	Enable, U1 pin 1	Low, under 1 V		
3 V on J2 pin 1	Enable	6S: high, near the output voltage. 3S: stays low, because 12 V is below the cutoff		
3 V removed	Enable	Low again within about a second	—	
3 V on J2 pin 4 instead	Enable	As for pin 1	—	
3 V on pin 1, 15 kΩ clipped across R12	Enable	Low within half a second: the board now sees a flat pack. High again when unclipped	—	

If not: if enable never goes high, recheck the reference (step 5) and U1 pin 6, then C4's polarity and D4.

Step 8 · Connect the comparator to the switch

From here the board decides for itself whether to stay on. Take the link out of J3 and use a momentary short or the button itself. Keep the 3 V on J2 pin 1.

Part	Value	Note
Q3	BC546B NPN, TO-92	Holds the gates on while enable is high
R6	47 kΩ	Q3's base
R7	100 kΩ	Q3's base to emitter. Its body stands over pad 2, the hole without the printed ring

Check

Condition	Measure at	Expect	3S	6S
Press about 1 s, release	Output	Stays on	—	
Press about 1 s, release	Output	Goes off on release: never stays on at 3S		—
A quick tap	Output	Doesn't stay on	—	
Remove the 3 V while on	Output	Off within about a second	—	
15 kΩ clipped across R12 while on	Output	Off within half a second	—	

After it turns off, wait for the output to fall below about 15 V before putting the 3 V back or unclipping R12. Until then the bank still powers the comparator and can turn the board back on. That's a bench effect; in the robot the carrier drains the bank at once.

If not: if it never stays on, look at Q3 turned round, or R6. If it stays on without the 3 V, the J3 link is still in or Q3 is shorted.

Step 9 · Full power

The 22 Ω leaves the pack lead and becomes the load. Do the voltage-drop check before J10 goes in, while its pads are free for the resistor.

Part	Value	Note
P1	14 AWG	Pack +, permanent
P3	14 AWG	Pack −, permanent
P4	16 AWG	The BMS's B−
P5	14 AWG	Charge inlet +
F1	20 A mini blade fuse	The service rating

Take the small load out of J10 and put the 22 Ω across J10's pins 1 and 3 instead. The pack now connects directly.

Check

Condition	Measure at	Expect	3S	6S
Switch on, 22 Ω across the output	One MOSFET, source pin to drain pin	6 mV or less. About 11 mV means one pair isn't being driven: look at R3 or R4	—	

On 6S the 22 Ω dissipates about 29 W. Keep it screwed to its aluminium plate, and on only as long as the reading takes.

Then fit, and connect the carrier

Part	Value	Note
J4–J9	JST-PH 4-pin vertical headers	The six motor outputs
J10	JST-XH 3-pin header	The carrier's feed. Take the 22 Ω off its pads first

Plug the carrier's feed into J10, and the hold lead from J2 into the carrier's J7. Take the 3 V off J2.

Check

Condition	Measure at	Expect	3S	6S
Button pressed about ½ s	The robot	Stays on: the firmware holds it	—	
Button pressed 2 s while running	The robot	Turns off	—	

What this doesn't prove

Without a scope, this sequence can't show the soft start's timing (about 130 ms at a third of an amp), how fast the switch lets go under load, or Q5's reset of C1. Running at the full 24 A waits for the robot's first drive.

The board, from above

Each pad is labelled with its net. Purple marks each diode, with the triangle on the anode and the bar on the cathode; the green one is the zener, D1. Red copper is the top layer and blue the bottom.

